第 8 章 多元函数微分学及其应用

主要内容

平面点集和区域

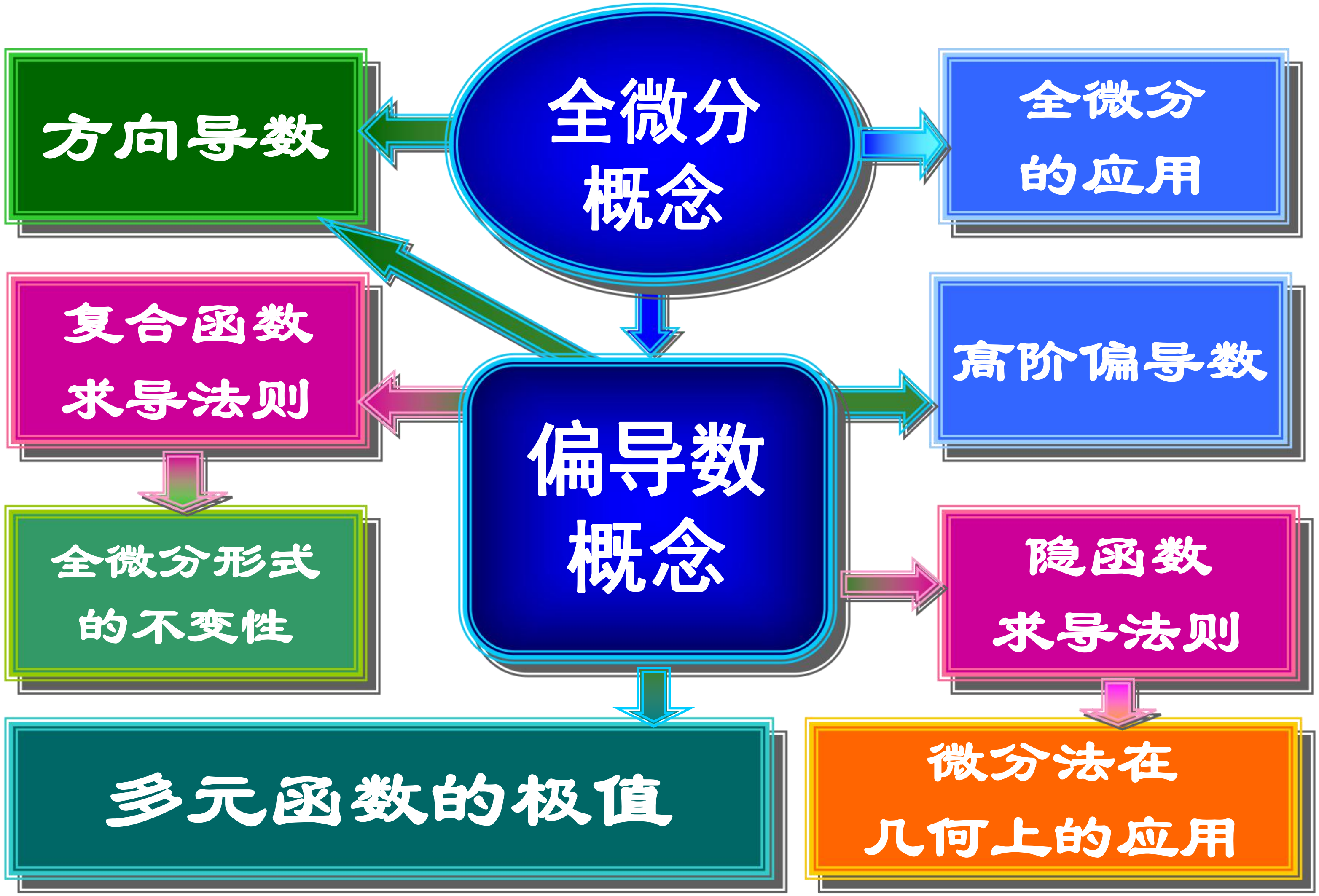
多元函数概念

多元函数的极限

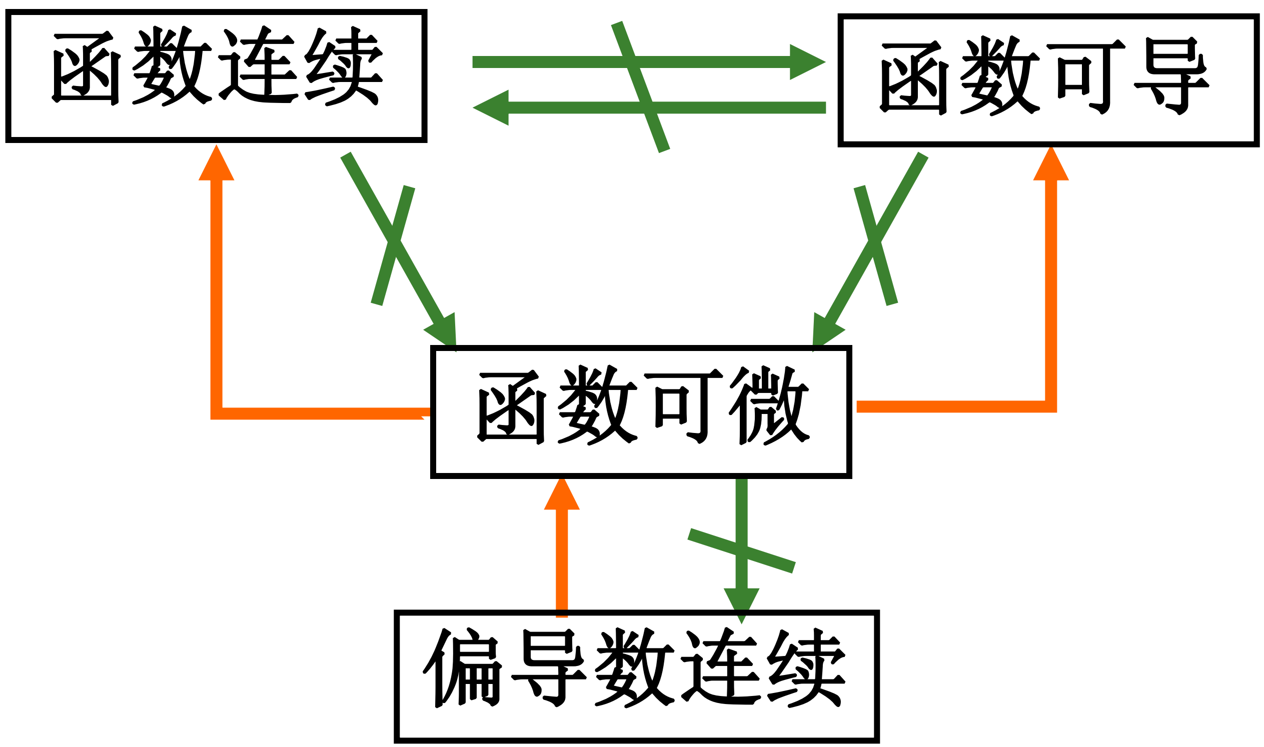
极 限 运 算

多元函数连续的概念

多元连续函数的性质


全微分概念
方向导数
全微分的应用
复合函数求导法则
偏导数概念
高阶偏导数
全微分形式的不变性
隐函数求导法则
多元函数的极值
微分法在几何上的应用

多元函数连续、可导、可微的关系.



练习1 设$z=f(x^2-y^2,xy)$,求$\dfrac{\partial^2 z}{\partial x^2}$、$\dfrac{\partial^2 z}{\partial x\partial y}$

其中$f(u,v)$的二阶偏导数连续.

练习2 求由方程 $xyz+\sqrt{x^2+y^2+z^2}=\sqrt{2}$

所确定的函数$z=z(x，y)$在点（1，0，-1）处的全微分.

练习1 设$z=f(x^2-y^2,xy)$,求$\dfrac{\partial^2 z}{\partial x^2}$、$\dfrac{\partial^2 z}{\partial x\partial y}$

其中$f(u,v)$的二阶偏导数连续.

$$u=x^2-y^2, v=xy$$

$$\frac{\partial z}{\partial x}=f_1'\cdot 2x+f_2'\cdot y \qquad \frac{\partial z}{\partial y}=f_1'\cdot(-2y)+f_2'\cdot x$$

$$\frac{\partial^2 z}{\partial x^2}=2f_1'+4x^2 f_{11}''+4xyf_{12}''+y^2 f_{22}''$$

$$\frac{\partial^2 z}{\partial x\partial y}=f_2'-4xyf_{11}''-2(y^2-x^2)f_{12}''+xyf_{22}''$$

练习2 求由方程 $xyz+\sqrt{x^2+y^2+z^2}=\sqrt{2}$

所确定的函数$z=z(x，y)$在点（1，0，-1）处的全微分.

解 将方程两边求全微分得

$$yzdx+xzdy+xydz+\frac{1}{\sqrt{x^2+y^2+z^2}}(xdx+ydy+zdz)=0,$$

因此，在点（1，0，－1）处

$$dz=dx-\sqrt{2}dy$$

举　　例

例1　设 $z = x^3 f(xy, \frac{y}{x})$, ($f$ 具有二阶连续偏导数),

求 $\frac{\partial z}{\partial y}, \frac{\partial^2 z}{\partial y^2}, \frac{\partial^2 z}{\partial x \partial y}$.

解　$$\frac{\partial z}{\partial y} = x^3 (f_1' x + f_2' \frac{1}{x}) = x^4 f_1' + x^2 f_2',$$

$$\frac{\partial^2 z}{\partial y^2} = x^4 (f_{11}'' x + f_{12}'' \frac{1}{x}) + x^2 (f_{21}'' x + f_{22}'' \frac{1}{x})$$

$$= x^5 f_{11}'' + 2x^3 f_{12}'' + x f_{22}'',$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial x}(x^4 f_1' + x^2 f_2')$$

$$= 4x^3 f_1' + x^4[f_{11}'' y + f_{12}''(-\frac{y}{x^2})] + 2xf_2'$$

$$+ x^2[f_{21}'' y + f_{22}''(-\frac{y}{x^2})]$$

$$= 4x^3 f_1' + 2xf_2' + x^4 y f_{11}'' - y f_{22}''.$$

例2. 设 $z = xf(x+y)$, $F(x,y,z)=0$, 其中 f 与F分别具有一阶导数或偏导数, 求 $\dfrac{\mathrm{d}z}{\mathrm{d}x}$. (99 考研)

解法1 方程两边对 x 求导, 得

$$\begin{cases} \dfrac{\mathrm{d}z}{\mathrm{d}x} = f + xf' \cdot \left(1+\dfrac{\mathrm{d}y}{\mathrm{d}x}\right) \\ F_1' + F_2'\dfrac{\mathrm{d}y}{\mathrm{d}x} + F_3'\dfrac{\mathrm{d}z}{\mathrm{d}x} = 0 \end{cases} \longrightarrow \begin{cases} -xf'\dfrac{\mathrm{d}y}{\mathrm{d}x} + \dfrac{\mathrm{d}z}{\mathrm{d}x} = f + xf' \\ F_2'\dfrac{\mathrm{d}y}{\mathrm{d}x} + F_3'\dfrac{\mathrm{d}z}{\mathrm{d}x} = -F_1' \end{cases}$$

$$\therefore \frac{\mathrm{d}z}{\mathrm{d}x} = \frac{\begin{vmatrix} -xf' & f + xf' \\ F_2' & -F_1' \end{vmatrix}}{\begin{vmatrix} -xf' & 1 \\ F_2' & F_3' \end{vmatrix}} = \frac{xF_1'f' - xF_2'f' - fF_2'}{-xf'F_3' - F_2'}$$

$(xf'F_3' + F_2' \neq 0)$

$$z = xf(x+y)\ ,\ F(x,y,z) = 0$$

解法2 方程两边求微分, 得

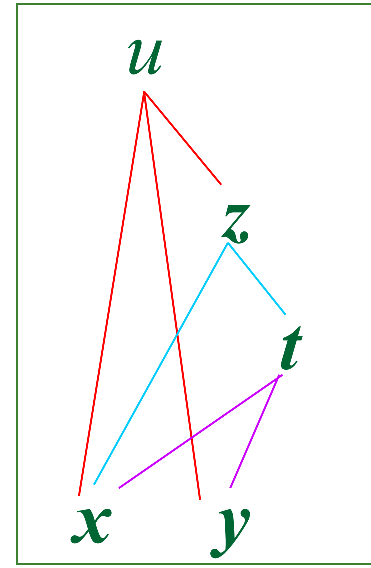
$$\begin{cases} \mathrm{d}z = f\,\mathrm{d}x + xf'\cdot(\mathrm{d}x + \mathrm{d}y) \\ F_1'\mathrm{d}x + F_2'\mathrm{d}y + F_3'\mathrm{d}z = 0 \end{cases}$$

$$\Big\downarrow \text{化简}$$

$$\begin{cases} (f + xf')\mathrm{d}x + xf\,\mathrm{d}y - \mathrm{d}z = 0 \\ F_1'\,\mathrm{d}x + F_2'\,\mathrm{d}y + F_3'\mathrm{d}z = 0 \end{cases}$$

消去 $\mathrm{d}y$ 即可得 $\dfrac{\mathrm{d}z}{\mathrm{d}x}$.

例3. 设 $u=f(x,y,z)$ 有二阶连续偏导数, 且 $z=x^2\sin t$, $t=\ln(x+y)$ 求 $\dfrac{\partial u}{\partial x}$, $\dfrac{\partial^2 u}{\partial x\partial y}$.



解: $$\frac{\partial u}{\partial x}=f_1'+f_3'\cdot\left(2x\sin t+x^2\cos t\cdot\frac{1}{x+y}\right)$$

$$\frac{\partial^2 u}{\partial x\partial y}=f_{12}''+f_{13}''\cdot\left(x^2\cos t\cdot\frac{1}{x+y}\right)$$

$$+\left[f_{32}''+f_{33}''\cdot\left(x^2\cos t\cdot\frac{1}{x+y}\right)\right]\left(2x\sin t+\frac{x^2\cos t}{x+y}\right)$$

$$+f_3'\cdot\left[2x\cos t\cdot\frac{1}{x+y}+x^2\frac{-\sin t\cdot\frac{1}{x+y}\,(x+y)-\cos t\cdot 1}{(x+y)^2}\right]$$

$$=\cdots$$

例4 已知：$x+y-z=e^z, xe^x=\tan t, y=\cos t$，求$\left.\dfrac{d^2z}{dt^2}\right|_{t=0}$

解 $\begin{cases} x+y-z-e^z=0 \\ xe^x-\tan t=0 \\ y-\cos t=0 \end{cases} \Rightarrow \begin{cases} x'+y'-z'-e^z z'=0 \\ e^x(1+x)x'-\sec^2 t=0 \\ y'+\sin t=0 \end{cases}$

$t=0 \Rightarrow x=0, y=1, z=0$ $\quad y'=0, x'=1, z'=\dfrac{1}{2}$

$$\Rightarrow \begin{cases} x''+y''-z''-e^z(z')^2-e^z z''=0 \\ e^x(2+x)x'^2+e^x(1+x)x''-2\sec^2 t\tan t=0 \\ y''+\cos t=0 \end{cases}$$

$y''=-1, x''=-2, \; -3-z''-\dfrac{1}{4}-z''=0 \quad \Rightarrow \left.z''\right|_{t=0}=-\dfrac{13}{8}$

例5设变换 $\begin{cases} u = x - 2y \\ v = x + ay \end{cases}$ 可把方程 $6\dfrac{\partial^2 z}{\partial x^2} + \dfrac{\partial^2 z}{\partial x \partial y} - \dfrac{\partial^2 z}{\partial y^2} = 0$

简化为 $\dfrac{\partial^2 z}{\partial u \partial v} = 0$，求常数$a$.

解 $\dfrac{\partial z}{\partial x} = \dfrac{\partial z}{\partial u} + \dfrac{\partial z}{\partial v}$， $\dfrac{\partial z}{\partial y} = -2\dfrac{\partial z}{\partial u} + a\dfrac{\partial z}{\partial v}$，

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial^2 z}{\partial u^2} + 2\frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2}, \quad \frac{\partial^2 z}{\partial y^2} = 4\frac{\partial^2 z}{\partial u^2} - 4a\frac{\partial^2 z}{\partial u \partial v} + a^2\frac{\partial^2 z}{\partial v^2},$$

$$\frac{\partial^2 z}{\partial x \partial y} = -2\frac{\partial^2 z}{\partial u^2} + (a-2)\frac{\partial^2 z}{\partial u \partial v} + a\frac{\partial^2 z}{\partial v^2}.$$

将上述结果代入原方程，经整理后得

$$(10+5a)\frac{\partial^2 z}{\partial u \partial v} + (6+a-a^2)\frac{\partial^2 z}{\partial v^2} = 0.$$

依题意a应满足 $6 + a - a^2 = 0$ 且 $10 + 5a \neq 0$ 解之得 a=3.

例6 设 $u = f(x,y,z), \varphi(x^2, e^y, z) = 0, y = \sin x,$

$(f,\varphi$ 具有一阶连续偏导数), 且 $\dfrac{\partial\varphi}{\partial z} \neq 0$, 求 $\dfrac{du}{dx}$.

解 $$\frac{du}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}\cdot\frac{dy}{dx} + \frac{\partial f}{\partial z}\frac{dz}{dx},$$

显然 $$\frac{dy}{dx} = \cos x,$$

求 $\dfrac{dz}{dx}$, 对 $\varphi(x^2, e^y, z) = 0$ 两边求 x 的导数,得

$$\varphi_1'\cdot 2x + \varphi_2'\cdot e^y\frac{dy}{dx} + \varphi_3'\frac{dz}{dx} = 0,$$

于是可得，$\dfrac{dz}{dx} = -\dfrac{1}{\varphi_3'}(2x\varphi_1' + e^{\sin x}\cdot\cos x\varphi_2')$,

故 $\dfrac{du}{dx} = \dfrac{\partial f}{\partial x} + \cos x\dfrac{\partial f}{\partial y} - \dfrac{1}{\varphi_3'}(2x\varphi_1' + e^{\sin x}\cdot\cos x\varphi_2')\dfrac{\partial f}{\partial z}$.

例7 设函数 $u(x)$ 由方程组 $\begin{cases} u = f(x,y), \\ g(x,y,z) = 0, \\ h(x,z) = 0. \end{cases}$

所确定,且 $\dfrac{\partial g}{\partial y} \neq 0, \dfrac{\partial h}{\partial z} \neq 0$, 试求 $\dfrac{du}{dx}$.

解 将方程组的变元 u 以及 y,z 都看成是 x 的函数.

方程组各方程两边对 x 求导,得

$$\begin{cases} \dfrac{du}{dx} = f_x + f_y \dfrac{dy}{dx}, & (1) \\ g_x + g_y \cdot \dfrac{dy}{dx} + g_z \cdot \dfrac{dz}{dx} = 0, & (2) \\ h_x + h_z \cdot \dfrac{dz}{dx} = 0. & (3) \end{cases}$$

由(3)得 $\dfrac{dz}{dx} = -\dfrac{h_x}{h_z}$, 代入(2)得 $\dfrac{dy}{dx} = \dfrac{g_z \cdot h_x}{g_y \cdot h_z} - \dfrac{g_x}{g_y}$,

代入(1)得 $\dfrac{du}{dx} = f_x - \dfrac{f_y \cdot g_x}{g_y} + \dfrac{f_y \cdot g_z \cdot h_x}{g_y \cdot h_z}$.

例8 求 $u=\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}$ 在点 $M(x_0,y_0,z_0)$ 处沿点的向径 r_0 的方向导数，问 a,b,c 具有什么关系时此方向导数等于梯度的模?

解 $\because \quad r_0=\{x_0,y_0,z_0\}, \quad |r_0|=\sqrt{x_0^2+y_0^2+z_0^2},$

$$\cos\alpha=\frac{x_0}{|r_0|}, \quad \cos\beta=\frac{y_0}{|r_0|}, \quad \cos\gamma=\frac{z_0}{|r_0|}.$$

$\therefore$ 在点 M 处的方向导数为

$$\frac{\partial u}{\partial r_0}\Big|_M=\frac{\partial u}{\partial x}\Big|_M\cos\alpha+\frac{\partial u}{\partial y}\Big|_M\cos\beta+\frac{\partial u}{\partial z}\Big|_M\cos\gamma$$

$$= \frac{2x_0}{a^2}\frac{x_0}{|r_0|} + \frac{2y_0}{b^2}\frac{y_0}{|r_0|} + \frac{2z_0}{c^2}\frac{z_0}{|r_0|} = \frac{2}{|r_0|}\left(\frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} + \frac{z_0^2}{c^2}\right)$$

$$= \frac{2u(x_0, y_0, z_0)}{\sqrt{x_0^2 + y_0^2 + z_0^2}}.$$

$\therefore$ 在点 M 处的梯度为

$$gradu\big|_M = \frac{\partial u}{\partial x}\Big|_M i + \frac{\partial u}{\partial y}\Big|_M j + \frac{\partial u}{\partial z}\Big|_M k$$

$$= \frac{2x_0}{a^2} i + \frac{2y_0}{b^2} j + \frac{2z_0}{c^2} k,$$

$$\left|gradu\right|_M = 2\sqrt{\frac{x_0^2}{a^4}+\frac{y_0^2}{b^4}+\frac{z_0^2}{c^4}},$$

当 $|a|=|b|=|c|$ 时, $\because$ $\left|gradu\right|_M = \dfrac{2}{a^2}\sqrt{x_0^2+y_0^2+z_0^2},$

$$\left.\frac{\partial u}{\partial r_0}\right|_M = \frac{\frac{2}{a^2}(x_0^2+y_0^2+z_0^2)}{\sqrt{x_0^2+y_0^2+z_0^2}} = \frac{2}{a^2}\sqrt{x_0^2+y_0^2+z_0^2},$$

$$\therefore \quad \left.\frac{\partial u}{\partial r_0}\right|_M = \left|gradu\right|_M,$$

故当 a,b,c 相等时,此方向导数等于梯度的 模.

课堂练习

1. 设函数 f 二阶连续可微, 求下列函数的二阶偏导数 $\dfrac{\partial^2 z}{\partial x \partial y}$.

$$(1)\quad z = xf\left(\frac{y^2}{x}\right)$$

$$(2)\quad z = f\left(x + \frac{y^2}{x}\right)$$

$$(3)\quad z = f\left(x, \frac{y^2}{x}\right)$$

2. 设 $x = e^u \cos v,\ y = e^u \sin v,\ z = uv$, 求 $\dfrac{\partial z}{\partial x}, \dfrac{\partial z}{\partial y}$

3. 设 $u=f(x,y,z)$有连续的一阶偏导数 ,又函数 $y=y(x)$及 $z=z(x)$ 分别由下两式确定

$$e^{xy}-xy=2,\ e^{x}=\int_{0}^{x-z}\frac{\sin t}{t}\,dt$$ 求 $\dfrac{\mathrm{d}u}{\mathrm{d}x}$. (2001考研)

4. 设$y=f(x,t)$, 而t是由方程$F(x,y,t)=0$所确定的 x,y的函数，其中 f,F具有一阶连续偏导数 ,证明

$$\frac{dy}{dx}=\frac{\dfrac{\partial f}{\partial x}\dfrac{\partial F}{\partial t}-\dfrac{\partial f}{\partial t}\dfrac{\partial F}{\partial x}}{\dfrac{\partial f}{\partial t}\dfrac{\partial F}{\partial y}+\dfrac{\partial F}{\partial t}}.$$

解答提示：第 1 题

(1) $z = xf(\frac{y^2}{x})$:
$$\frac{\partial z}{\partial y} = xf'(\frac{y^2}{x}) \cdot \frac{2y}{x} = 2yf'$$

$$\frac{\partial^2 z}{\partial x \partial y} = 2yf'' \cdot (-\frac{y^2}{x^2}) = -\frac{2y^3}{x^2} f''$$

(2) $z = f(x + \frac{y^2}{x})$:
$$\frac{\partial z}{\partial y} = f' \cdot \frac{2y}{x}$$

$$\frac{\partial^2 z}{\partial x \partial y} = -\frac{2y}{x^2} f' + \frac{2y}{x} f'' \cdot (1 - \frac{y^2}{x^2})$$

$$= -\frac{2y}{x^2} f' + \frac{2y}{x}(1 - \frac{y^2}{x^2}) f''$$

(3) $z = f(x, \dfrac{y^2}{x})$:

$$\frac{\partial z}{\partial y} = \frac{2y}{x} f_2'$$

$$\frac{\partial^2 z}{\partial x \partial y} = -\frac{2y}{x^2} f_2' + \frac{2y}{x}\left(f_{21}'' - \frac{y^2}{x^2} f_{22}'' \right)$$

2. 设 $x=e^u\cos v,\ y=e^u\sin v,\ z=uv$, **求** $\dfrac{\partial z}{\partial x},\dfrac{\partial z}{\partial y}$

解 由 $z=uv$, 得

$$\frac{\partial z}{\partial x}=v\frac{\partial u}{\partial x}+u\frac{\partial v}{\partial x} \qquad ①$$

$$\frac{\partial z}{\partial y}=v\frac{\partial u}{\partial y}+u\frac{\partial v}{\partial y} \qquad ②$$

由 $x=e^u\cos v,\ \ y=e^u\sin v$, 得

$$\begin{cases}\mathrm{d}x=e^u\cos v\,\mathrm{d}u-e^u\sin v\,\mathrm{d}v\\ \mathrm{d}y=e^u\sin v\,\mathrm{d}u+e^u\cos v\,\mathrm{d}v\end{cases}$$

利用行列式解出 du, dv :

$$\mathrm{d}u = \frac{\begin{vmatrix} \mathrm{d}x & -e^u \sin v \\ \mathrm{d}y & e^u \cos v \end{vmatrix}}{\begin{vmatrix} e^u \cos v & -e^u \sin v \\ e^u \sin v & e^u \cos v \end{vmatrix}} = \underbrace{e^{-u} \cos v}_{\frac{\partial u}{\partial x}} \mathrm{d}x + \underbrace{e^{-u} \sin v}_{\frac{\partial u}{\partial y}} \mathrm{d}y$$

$$\mathrm{d}v = -\underbrace{e^{-u} \sin v}_{\frac{\partial v}{\partial x}} \mathrm{d}x + \underbrace{e^{-u} \cos v}_{\frac{\partial v}{\partial y}} \mathrm{d}y$$

将 $\frac{\partial u}{\partial x}$ 及 $\frac{\partial v}{\partial x}$ 代入①即得 $\frac{\partial z}{\partial x}$;

将 $\frac{\partial u}{\partial y}$ 及 $\frac{\partial v}{\partial y}$ 代入②即得 $\frac{\partial z}{\partial y}$.

3. 设 $u=f(x,y,z)$ 有连续的一阶偏导数，又函数 $y=y(x)$ 及 $z=z(x)$ 分别由下两式确定

$$e^{xy}-xy=2,\quad e^{x}=\int_0^{x-z}\frac{\sin t}{t}\,dt$$ 求 $\frac{\mathrm{d}u}{\mathrm{d}x}$.（2001考研）

解: 两个隐函数方程两边对 x 求导, 得

$$\begin{cases} e^{xy}(y+xy')-(y+xy')=0 \\ e^{x}=\dfrac{\sin(x-z)}{x-z}(1-z') \end{cases}$$

解得 $$y'=-\frac{y}{x},\quad z'=1-\frac{e^{x}(x-z)}{\sin(x-z)}$$

因此 $$\frac{\mathrm{d}u}{\mathrm{d}x}=f_1'-\frac{y}{x}f_2'+\left[\ 1-\frac{e^{x}(x-z)}{\sin(x-z)}\ \right]f_3'$$

4. 设$y=f(x,t)$，而t是由方程$F(x,y,t)=0$所确定的x,y的函数，其中f,F具有一阶连续偏导数,证明

$$\frac{dy}{dx}=\frac{\frac{\partial f}{\partial x}\frac{\partial F}{\partial t}-\frac{\partial f}{\partial t}\frac{\partial F}{\partial x}}{\frac{\partial f}{\partial t}\frac{\partial F}{\partial y}+\frac{\partial F}{\partial t}}.$$

解法1

$$\begin{cases} y-f\left(x,t\right)=0 \\ F\left(x,y,t\right)=0 \end{cases} \Rightarrow \begin{matrix} t=t\left(x\right) \\ y=y\left(x\right) \end{matrix}$$

将上式两方程对x求导数：

$$\begin{cases} \dfrac{dy}{dx}-f_x'-f_t'\dfrac{dt}{dx}=0 \\ F_x'+F_y'\dfrac{dy}{dx}+F_t'\dfrac{dt}{dx}=0 \end{cases},$$

$$\frac{dy}{dx}=\begin{vmatrix} f_x' & -f_t' \\ -F_x' & F_t' \end{vmatrix}\Big/\begin{vmatrix} 1 & -f_t' \\ F_y' & F_t' \end{vmatrix}=\frac{f_x'F'_t-f_t'F_x'}{F_t'+f_t'F_y'}$$

解法2 $dy = f_x' dx + f_t' dt \quad (1)$

$$F_x dx + F_y dy + F_t dt = 0 \quad (2)$$

由（2） $dt = -\dfrac{F_x dx + F_y dy}{F_t}$ 代入(1)得

$$dy = f_x' dx + f_t' \left(-\frac{F_x dx + F_y dy}{F_t}\right)$$

$$dy = \frac{f_x F_t - f_t F_x}{F_t + f_t F_y} dx \qquad \therefore \frac{dy}{dx} = \frac{f_x F_t - f_t F_x}{F_t + f_t F_y}$$

$$\begin{cases} y = f(x,t) \\ F(x,y,t) = 0 \end{cases} \Rightarrow \begin{matrix} t = t(x) \\ y = y(x) \end{matrix}$$

EX

1. 已知 $f(x,x^2)=1$，$f_1'(x,x^2)=2x$，求 $f_2'(x,x^2)$.

2. 设函数 $z=f(x,y)$ 在点$(1,1)$处可微，且

$$f(1,1)=1,\quad \left.\frac{\partial f}{\partial x}\right|_{(1,1)}=2,\quad \left.\frac{\partial f}{\partial y}\right|_{(1,1)}=3,$$

$\varphi(x)=f(\underline{x},\underline{f(x,x)})$，求 $\left.\dfrac{\mathrm{d}}{\mathrm{d}x}\varphi^3(x)\right|_{x=1}$. **(2001考研)**

EX1

1. 已知 $f(x,x^2)=1$，$f_1'(x,x^2)=2x$，求 $f_2'(x,x^2)$.

解 由 $f(x,x^2)=1$ 两边对 x 求导，得

$$f_1'(x,x^2)+f_2'(x,x^2)\cdot 2x=0$$

$$\Big\downarrow\; f_1'(x,x^2)=2x$$

$$f_2'(x,x^2)=-1$$

2. 设函数 $z=f(x,y)$ 在点$(1,1)$处可微，且

$$f(1,1)=1,\quad \left.\frac{\partial f}{\partial x}\right|_{(1,1)}=2,\quad \left.\frac{\partial f}{\partial y}\right|_{(1,1)}=3,$$

$\varphi(x)=f(x,f(x,x))$，求 $\left.\frac{\mathrm{d}}{\mathrm{d}x}\varphi^3(x)\right|_{x=1}$. **(2001考研)**

解: 由题设 $\varphi(1)=f(1,f(1,1))=f(1,1)=1$

$$\left.\frac{\mathrm{d}}{\mathrm{d}x}\varphi^3(x)\right|_{x=1}=3\varphi^2(x)\left.\frac{\mathrm{d}\varphi}{\mathrm{d}x}\right|_{x=1}$$

$$=3\Big[f_1'(x,f(x,x))$$

$$\left.+f_2'(x,f(x,x))\big(f_1'(x,x)+f_2'(x,x)\big)\Big]\right|_{x=1}$$

$$=3\cdot\left[2+3\cdot(2+3)\right]=51$$

(1).分清哪些是中间变量， 哪些是自变量，及其相互关系

(2).分清偏导与全导记号

(3).抽象函数常采用简便记 号：

$f_1', f_2', f''_{11}, f''_{12}$ 等

(4). 对于高阶偏导数

i)运用四则运算法则，把一阶偏导数拆开

ii)记住f_1', f_2'仍与f的复合关系一样

隐函数微分法

(1)由$F(x,y)=0$确定隐函数$y=y(x)$则

有$y'(x)=-\dfrac{F_x(x,y)}{F_y(x,y)}$,$(F_y(x,y)\neq 0)$

(2)由$F(x,y,z)=0$确定隐函数$z=z(x,y)$则

$$\frac{\partial z}{\partial x}=-\frac{F_x(x,y,z)}{F_z(x,y,z)},\ \frac{\partial z}{\partial y}=-\frac{F_y(x,y,z)}{F_z(x,y,z)}$$

$(F_z(x,y,z)\neq 0)$

(3)由多个变量确定的隐函 数组微分法

—不用记公式,直接对方程两边求导数

(或偏导数)这种方法称为直接法

（相对于公式）

求一阶时，可用一阶全微分形式不变性.

求二阶或二阶以上（偏）导数,

通常用直接法

多元函数微分法的应用

1.几何应用

(1) 空间曲线的切线与法平面

$$\Gamma: x = \varphi(t),\ y = \psi(t),\ z = \omega(t).$$

切线方程为 $\dfrac{x - x_0}{\varphi'(t_0)} = \dfrac{y - y_0}{\psi'(t_0)} = \dfrac{z - z_0}{\omega'(t_0)}.$

$$\vec{\tau} = \{\varphi'(t_0), \psi'(t_0), \omega'(t_0)\}.$$

法平面方程为

$$\varphi'(t_0)(x - x_0) + \psi'(t_0)(y - y_0) + \omega'(t_0)(z - z_0) = 0.$$

$$\Gamma: \begin{cases} x = x, \\ y = y(t), \\ z = z(t) \end{cases}$$

$$\Gamma: \begin{cases} y = f(x) \\ z = 0 \end{cases}$$

$$\vec{\tau} = ?$$

（2） 曲面的切平面与法线

$$\pi: \quad F(x,y,z)=0.$$

$$\vec{n}=\{F_x(x_0,y_0,z_0),F_y(x_0,y_0,z_0),F_z(x_0,y_0,z_0)\}$$

切平面方程为

$$F_x(x_0,y_0,z_0)(x-x_0)$$
$$+F_y(x_0,y_0,z_0)(y-y_0)$$
$$+F_z(x_0,y_0,z_0)(z-z_0)=0$$

$$S: z=f(x,y)$$

$$\vec{n}=?$$

法线方程为

$$\frac{x-x_0}{F_x(x_0,y_0,z_0)}=\frac{y-y_0}{F_y(x_0,y_0,z_0)}=\frac{z-z_0}{F_z(x_0,y_0,z_0)}.$$

3. 多元函数的极值与最值

•*(无条件)极值* ——对自变量除限制在定义域内，没有其它限制。

(1)定义　　*(2) 驻点*

(3)判定

定理 若点(x_0, y_0)为驻点，

记，$\Delta = f_{xx}(x_0, y_0) f_{yy}(x_0, y_0) - (f_{xy}(x_0, y_0))^2$

$$\Rightarrow \begin{cases} (1)\text{当}\Delta > 0\text{时，}(x_0, y_0)\text{为极值点且} f_{xx} < 0 \text{极大点} \\ (2)\text{当}\Delta < 0\text{时，}(x_0, y_0)\text{不是极值点} \\ (3)\text{当}\Delta = 0\text{时，不能确定}(x_0, y_0)\text{是否为极值.} \end{cases}$$

> 小

• *条件极值*

求$z = f(x,y)$在附加条件 $\varphi(x,y) = 0$下的可能极值点

拉格朗日乘数法：

构造函数：$F(x,y) = f(x,y) + \lambda\varphi(x,y)$，（$\lambda$ 是参数）

从方程$\begin{cases} F_x = f_x(x,y) + \lambda\varphi_x(x,y) = 0 \\ F_y = f_y(x,y) + \lambda\varphi_y(x,y) = 0 \\ \varphi(x,y) = 0 \end{cases}$解出$x, y, \lambda$

$\Rightarrow (x,y)$就是可能的极值点.

•*求最值*

有界闭区域D上的多元连续函数必取得最大值最小值.

象一元函数那样将全部 驻点处的函数值求出与边界上的函数值比较 决定最大最小值 .

(1)二元函数求边界上的函 数值较麻烦 .

(2) 对实际问题,最值必存在,求出唯一的驻点,不加验证,就必是最值点.

举　　例

EX1　在曲线：$x=t, y=-t^2, z=t^3$上求一点使在该点的切线平行于平面

$$\pi: \ x+2y+z=0,$$

并求过该点的切线方程。

EX2 求曲面$x^2+2y^2+3z^2=6$上平行于直线$x=t, y=1+2t, z=-1+3t$的法线方程.

EX3　设$\vec{n}$是曲面$2x^2+3y^2+z^2=6$ 在点$P(1,1,1)$处的指向外侧的法向量，求函数

$u=\frac{1}{z}(6x^2+8y^2)^{\frac{1}{2}}$在 P 处沿方向$\vec{n}$的方向导数.

例1 在曲线：$x=t, y=-t^2, z=t^3$上求一点使在该点的切线平行于平面

$$\pi: \ x+2y+z=0,$$

并求过该点的切线方程。

解 设所求点对应的参数为t,

该点的切矢为: $\vec{\tau}=\{1,-2t,3t^2\}$

又π的法矢为$\vec{n}=\{1,2,1\}$,

依题意有$\vec{\tau}\cdot\vec{n}=\{1,-2t,3t^2\}\cdot\{1,2,1\}=0$,

即 $1-4t+3t^2=0$,解得$t_1=1, t_2=\dfrac{1}{3}$

$t_1=1$,得切点$(1,-1,1)$,切矢$\vec{\tau}_1=(1,-2,3)$

切线方程为:$\dfrac{x-1}{1}=\dfrac{y+1}{-2}=\dfrac{z-1}{3}$

$t_2=\dfrac{1}{3}$ 得切点$(\dfrac{1}{3},-\dfrac{1}{9},\dfrac{1}{27})$,切矢$\vec{\tau}_2=(1,-\dfrac{2}{3},\dfrac{1}{3})$

切线方程为:$\dfrac{x-1/3}{3}=\dfrac{y+1/9}{-2}=\dfrac{z-1/27}{1}$

例2 求曲面$x^2+2y^2+3z^2=6$上平行于直线

$x=t, y=1+2t, z=-1+3t$的法线方程.

解 设切点为(x_0, y_0, z_0),

令$F(x,y,z)=x^2+2y^2+3z^2-6$

则法线的方向向量为$\vec{n}=\{2x_0, 4y_0, 6z_0\}$

直线的方向向量为$\{1,2,3\}$

依题意有$\dfrac{x_0}{1}=\dfrac{2y_0}{2}=\dfrac{3z_0}{3}$, $x_0{}^2+2y_0{}^2+3z_0{}^2=6$

解得$x_0=1, y_0=1, z_0=1; x_0=-1, y_0=-1, z_0=-1$

法线方程为:$\dfrac{x\pm1}{1}=\dfrac{y\pm1}{2}=\dfrac{z\pm1}{3}$

例 3 设$\vec{n}$是曲面$2x^2+3y^2+z^2=6$在点$P(1,1,1)$处的指向外侧的法向量，求函数$u=\frac{1}{z}(6x^2+8y^2)^{\frac{1}{2}}$在$P$处沿方向$\vec{n}$的方向导数.

解 令 $F(x,y,z)=2x^2+3y^2+z^2-6$,

$$F_x\big|_P=4x\big|_P=4,\quad F_y\big|_P=6y\big|_P=6,\quad F_z\big|_P=2z\big|_P=2,$$

故 $\vec{n}=(F_x,F_y,F_z)\big|_P=(4,6,2)$,

$|\vec{n}|=\sqrt{4^2+6^2+2^2}=2\sqrt{14}$, 方向余弦为

$$\cos\alpha = \frac{2}{\sqrt{14}}, \quad \cos\beta = \frac{3}{\sqrt{14}}, \quad \cos\gamma = \frac{1}{\sqrt{14}}.$$

$$\left.\frac{\partial u}{\partial x}\right|_P = \left.\frac{6x}{z\sqrt{6x^2+8y^2}}\right|_P = \frac{6}{\sqrt{14}};$$

$$\left.\frac{\partial u}{\partial y}\right|_P = \left.\frac{8y}{z\sqrt{6x^2+8y^2}}\right|_P = \frac{8}{\sqrt{14}};$$

$$\left.\frac{\partial u}{\partial z}\right|_P = -\left.\frac{\sqrt{6x^2+8y^2}}{z^2}\right|_P = -\sqrt{14}.$$

故 $\left.\frac{\partial u}{\partial \vec{n}}\right|_P = \left(\frac{\partial u}{\partial x}\cos\alpha + \frac{\partial u}{\partial y}\cos\beta + \frac{\partial u}{\partial z}\cos\gamma\right)\Big|_P = \frac{11}{7}.$

例4 在第一卦限内作椭球面 $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$ 的切平面，使切平面与三个坐标面所围成的四面体体积最小，求切点坐标.

解 设 $P(x_0, y_0, z_0)$ 为椭球面上一点,

令 $F(x,y,z)=\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}-1$,

则 $F'_x|_P=\frac{2x_0}{a^2}$， $F'_y|_P=\frac{2y_0}{b^2}$， $F'_z|_P=\frac{2z_0}{c^2}$

过 $P(x_0, y_0, z_0)$ 的切平面方程为

$$\frac{x_0}{a^2}(x-x_0)+\frac{y_0}{b^2}(y-y_0)+\frac{z_0}{c^2}(z-z_0)=0,$$

化简为 $\dfrac{x\cdot x_0}{a^2}+\dfrac{y\cdot y_0}{b^2}+\dfrac{z\cdot z_0}{c^2}=1,$

该切平面在三个轴上的截距各为

$$x=\frac{a^2}{x_0},\quad y=\frac{b^2}{y_0},\quad z=\frac{c^2}{z_0},$$

所围四面体的体积 $V=\dfrac{1}{6}xyz=\dfrac{a^2b^2c^2}{6x_0y_0z_0},$

在条件$\frac{x_0^2}{a^2}+\frac{y_0^2}{b^2}+\frac{z_0^2}{c^2}=1$下求 V 的最小值,

令 $u=\ln x_0+\ln y_0+\ln z_0$,

$$G(x_0,y_0,z_0)$$
$$=\ln x_0+\ln y_0+\ln z_0+\lambda(\frac{x_0^2}{a^2}+\frac{y_0^2}{b^2}+\frac{z_0^2}{c^2}-1),$$

由$\begin{cases} G'_{x_0}=0, \quad G'_{y_0}=0, \quad G'_{z_0}=0 \\ \frac{x_0^2}{a^2}+\frac{y_0^2}{b^2}+\frac{y_0^2}{c^2}-1=0 \end{cases}$,

即
$$\begin{cases} \dfrac{1}{x_0}+\dfrac{2\lambda x_0}{a^2}=0 \\ \dfrac{1}{y_0}+\dfrac{2\lambda y_0}{b^2}=0 \\ \dfrac{1}{z_0}+\dfrac{2\lambda z_0}{c^2}=0 \\ \dfrac{x_0^2}{a^2}+\dfrac{y_0^2}{b^2}+\dfrac{z_0^2}{c^2}-1=0 \end{cases}$$

可得
$$\begin{cases} x_0=\dfrac{a}{\sqrt{3}} \\ y_0=\dfrac{b}{\sqrt{3}}, \\ z_0=\dfrac{c}{\sqrt{3}} \end{cases}$$

当切点坐标为

$\left(\dfrac{a}{\sqrt{3}}, \dfrac{b}{\sqrt{3}}, \dfrac{c}{\sqrt{3}}\right)$时,

四面体的体积最小$V_{\min}=\dfrac{\sqrt{3}}{2}abc.$

在条件 $\dfrac{x_0^2}{a^2}+\dfrac{y_0^2}{b^2}+\dfrac{z_0^2}{c^2}=1$ 下求 $V=\dfrac{a^2b^2c^2}{6x_0y_0z_0}$ 的最小值

解

$$\left(\frac{x_0y_0z_0}{abc}\right)^2\le\left(\frac{\dfrac{x_0^2}{a^2}+\dfrac{y_0^2}{b^2}+\dfrac{z_0^2}{c^2}}{3}\right)^3=\frac{1}{27}$$

$$\Rightarrow\frac{x_0y_0z_0}{abc}\le\frac{1}{3\sqrt{3}}\Rightarrow\frac{abc}{x_0y_0z_0}\ge3\sqrt{3}\Rightarrow V_{\min}=\frac{\sqrt{3}}{2}abc$$

不等式等号成立 $\Leftrightarrow\dfrac{x_0}{a}=\dfrac{y_0}{b}=\dfrac{z_0}{c}$

代入 $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}+\dfrac{z^2}{c^2}=1$，得 $x_0=\dfrac{a}{\sqrt{3}},y_0=\dfrac{b}{\sqrt{3}},z_0=\dfrac{c}{\sqrt{3}}$

例5 求旋转抛物面 $z = x^2 + y^2$ 与平面 $x + y - 2z = 2$ 之间的最短距离.

解 设 $P(x, y, z)$ 为抛物面 $z = x^2 + y^2$ 上任一点,则 P 到平面 $x + y - 2z - 2 = 0$ 的距离为 d,

$$d = \frac{1}{\sqrt{6}}|x + y - 2z - 2|.$$

分析: 本题变为求一点 $P(x, y, z)$,使得 x, y, z 满足 $x^2 + y^2 - z = 0$且使 $d = \frac{1}{\sqrt{6}}|x + y - 2z - 2|$

(即 $d^2 = \frac{1}{6}(x + y - 2z - 2)^2$) 最小.

令 $F(x,y,z)=\frac{1}{6}(x+y-2z-2)^2+\lambda(z-x^2-y^2)$, 得

$$\begin{cases} F'_x=\frac{1}{3}(x+y-2z-2)-2\lambda x=0, & (1) \\ F'_y=\frac{1}{3}(x+y-2z-2)-2\lambda y=0, & (2) \\ F'_z=\frac{1}{3}(x+y-2z-2)(-2)+\lambda=0, & (3) \\ z=x^2+y^2, & (4) \end{cases}$$

解此方程组得 $x=\frac{1}{4}, y=\frac{1}{4}, z=\frac{1}{8}$.

即得唯一驻点 $(\frac{1}{4},\frac{1}{4},\frac{1}{8})$,

根据题意距离的最小值一定存在，且有唯一驻点，故必在 $(\frac{1}{4},\frac{1}{4},\frac{1}{8})$ 处取得最小值.

$$d_{\min}=\frac{1}{\sqrt{6}}\left|\frac{1}{4}+\frac{1}{4}-\frac{1}{4}-2\right|=\frac{7}{4\sqrt{6}}.$$

例6 设有两个正数x与y之和为定值a,求函数

$$f(x,y)=\frac{x^n+y^n}{2}$$的极小值,并证明:

$$\frac{x^n+y^n}{2}\ge\left(\frac{x+y}{2}\right)^n$$(n为正整数)

解 $$F(x,y,\lambda)=\frac{x^n+y^n}{2}+\lambda(x+y-a)$$

$$\begin{cases}\left.\begin{aligned}F_x&=\frac{n}{2}x^{n-1}+\lambda\overset{\text{令}}{=}0\\F_y&=\frac{n}{2}y^{n-1}+\lambda\overset{\text{令}}{=}0\end{aligned}\right\}\Rightarrow x=y\\x+y=a\qquad\qquad x=\frac{a}{2}=y\end{cases}$$

$$\therefore f_{\min}=\left(\frac{a}{2}\right)^n,$$

$$\therefore f(x,y)\ge\left(\frac{a}{2}\right)^n=\left(\frac{x+y}{2}\right)^n$$

课堂练习

一、选择题

1、曲面$xyz=a^3(a>0)$的切平面与三个坐标面所围成的四面体的体积 V=((C)).

(A) $\frac{3}{2}a^3$;　　(B) $3a^3$;

(C) $\frac{9}{2}a^3$;　　(D) $6a^3$.

2、二元函数$z=3(x+y)-x^3-y^3$的极值点是((D)).

(A) (1,4);　　(B) (1.-2);

(C) (-1,2);　　(D) (-1,-1).

3. 函数$z=2x-y$在由$x+y=1, y=x+1$及$y=0$所围闭域$\overline{D}$上最小值（(C)）

（A）0 （B）-1 （C）-2 （D）不∃

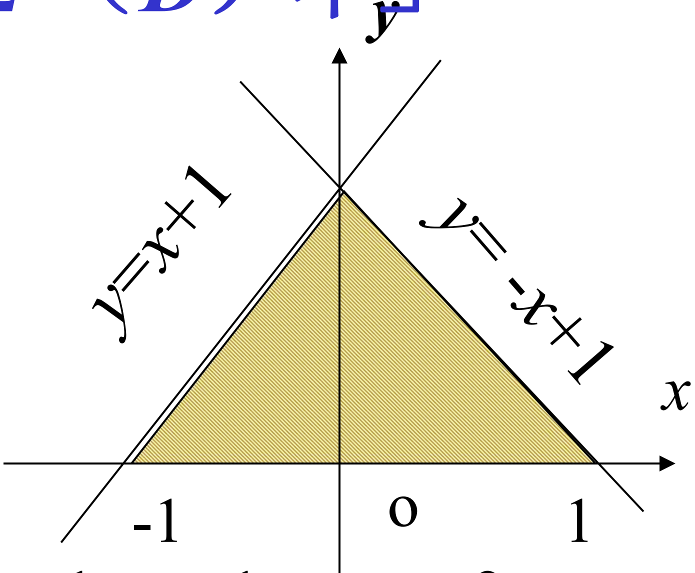
4. 函数$u=xyz$在点(1, 2，−2)增长最快的方向 $-4\vec{i}-2\vec{j}+2\vec{k}$ 和变化率 $2\sqrt{6}$.

3. 函数$z=2x-y$在由$x+y=1, y=x+1$及$y=0$所围闭域$\overline{D}$上最小值（(C)）

（A）0 （B）-1 （C）-2 （D）不∃

解 $z_x=2 \quad z_y=-1,$

$\therefore$ 最值在边界上达到。



在$y=0$上，$z=2x \quad -1\le x\le 1$

$z_{\min}=-2, z_{\max}=2$

在$y=x+1 \quad z=2x-(x+1)=x-1 \quad -1\le x\le 0$

$z_{\min}=-2, z_{\max}=-1$

在$y=1-x \quad 0\le x\le 1 \quad z=3x-1$

$z_{\min}=-1, z_{\max}=2 \quad \therefore$ 在$\overline{D}$上最小值为-2.

4.　函数$u=xyz$在点(1, 2, −2)增长最快的方向

$-4\vec{i}-2\vec{j}+2\vec{k}$ 和变化率 $2\sqrt{6}$.

$$\overrightarrow{\mathbf{grad}}u=(yz\ ,\ xz\ ,\ xy)\Big|_{(1,2,-2)}=(-4,-2,2)$$

$$\left|\overrightarrow{\mathbf{grad}}u\right|=2\sqrt{6}$$

二、求过直线$\begin{cases} x+2y+z-1=0 \\ x-y-2z+3=0 \end{cases}$的平面$\pi$,使之平行于曲线$\begin{cases} x^2+y^2=\frac{z^2}{2} \\ x+y+2z=4 \end{cases}$在点$(1,-1,2)$的切线.

三、已知正数a为3个正的因子的乘积,使它们的和为最小.

二、求过直线$\begin{cases} x+2y+z-1=0 \\ x-y-2z+3=0 \end{cases}$的平面$\pi$，使之平行于曲线$\begin{cases} x^2+y^2=\frac{z^2}{2} \\ x+y+2z=4 \end{cases}$在点$(1,-1,2)$的切线.

解 设所求平面方程 π为

$$x+2y+z-1+\lambda(x-y-2z+3)=0$$

该平面的法向量 $\vec{n}=\{(1+\lambda),(2-\lambda),(1-2\lambda)\}$

曲线$\begin{cases} x^2+y^2=\frac{z^2}{2} \\ x+y+2z=4 \end{cases}$在点$M_0(1,-1,2)$的切线的方向向量 $\vec{l}$

$$令 F(x,y,z) = x^2 + y^2 - \frac{z^2}{2} = 0$$

$$G(x,y,z) = x + y + 2z - 4 = 0$$

$$\vec{n}_1 = \{2x, 2y, -z\}_{M_0} = \{2,-2,-2\}, \vec{n}_2 = \{1,1,2\}$$

$$\vec{l} = \vec{n}_1 \times \vec{n}_2 = 2\{\begin{vmatrix} -1 & -1 \\ 1 & 2 \end{vmatrix} \begin{vmatrix} -1 & 1 \\ 2 & 1 \end{vmatrix} \begin{vmatrix} 1 & -1 \\ 1 & 1 \end{vmatrix}\}$$

$$\vec{l} = 2\{-1,-3,2\}$$

已知所求平面平行切线,所以 $\vec{n} \cdot \vec{l} = 0$

$$\{(1+\lambda),(2-\lambda),(1-2\lambda)\} \cdot \{-1,-3,2\} = 0$$

解得 $\lambda = -\frac{5}{2}$,因此所求平面 $3x - 9y - 12z + 17 = 0$

三、已知正数a为3个正的因子的乘积，使它们的和为最小.

解法一 问题就是在条件 $xyz = a$下

求$u = x + y + z$的最小值.

$$u = x + y + z \geq 3\sqrt[3]{xyz} = 3\sqrt[3]{a}$$

又当$x = y = z = \sqrt[3]{a}$时，

$u = 3\sqrt[3]{a}$故u的最小值为 $3\sqrt[3]{a}$.

解法二 $F(x,y,z)=x+y+z+\lambda(xyz-a)$

$$\begin{cases} F_x=1+\lambda yz\overset{令}{=}0 \\ F_y=1+\lambda xz\overset{令}{=}0 \\ F_z=1+\lambda xy\overset{令}{=}0 \\ xyz=a \end{cases} \Rightarrow x=y=z$$

代入$xyz=a$得$x=\sqrt[3]{a}$

解法三 化为无条件极值 $,u=x+y+\dfrac{a}{xy}$

四、求曲线$\begin{cases} 2x-3y+z=0 \\ 2x^2+3y^2+z^2=30 \end{cases}$

上竖坐标的最大最小值

五、设$F(u,v)$可微，试证曲面$S:F(\dfrac{x-a}{z-c},\dfrac{y-b}{z-c})=0$

上任一点的切平面都通过定点。

四、求曲线$\begin{cases} 2x-3y+z=0 \\ 2x^2+3y^2+z^2=30 \end{cases}$上竖坐标的最大最小值

解

令$F=z+\lambda(2x-3y+z)+\mu(2x^2+3y^2+z^2-30)$

$$\begin{cases} F_x=2\lambda+4\mu x=0 \\ F_y=-3\lambda+6\mu y=0 \quad \Rightarrow x=-y \\ F_z=1+\lambda+2\mu z=0 \\ 2x-3y+z=0 \\ 2x^2+3y^2+z^2=30 \quad \Rightarrow (\pm 1, \mp 1, \mp 5) \end{cases}$$

$$\therefore z_{\max}=5 \quad z_{\min}=-5.$$

五、设$F(u,v)$可微，试证曲面$S: F(\frac{x-a}{z-c}, \frac{y-b}{z-c})=0$上任一点的切平面都通过定点。

证明：

$\forall(x_0,y_0,z_0)\in S$ 该点的切平面法向量：

$$\vec{n}=\{F_1'\cdot\frac{1}{z_0-c},F_2'\cdot\frac{1}{z_0-c},-[F_1'\cdot\frac{x_0-a}{(z_0-c)^2}+F_2'\cdot\frac{y_0-b}{(z_0-c)^2}]\}$$

切平面方程π：

$$F_1'\cdot\frac{1}{z_0-c}(x-x_0)+F_2'\cdot\frac{1}{z_0-c}(y-y_0)$$

$$-[F_1'\cdot\frac{x_0-a}{(z_0-c)^2}+F_2'\cdot\frac{y_0-b}{(z_0-c)^2}](z-z_0)=0$$

易见$(x,y,z)=(a,b,c)\in\pi$

故曲面S在任何点处的切平面都 通过定点$p_0(a,b,c)$

六 设函数 $f(x,y,z)=x^2+y^z$

(1) 求函数在点 $M(1,1,1)$ 处沿曲线 $\begin{cases} x=t \\ y=2t^2-1 \\ z=t^3 \end{cases}$ 在该点切线方向的方向导数;

(2) 求函数在 $M(1,1,1)$ 处的梯度与(1)中切线方向的夹角 θ.

解答提示:

1\. (1) $f(x,y,z)=x^2+y^z$, 曲线 $\begin{cases} x=t \\ y=2t^2-1 \\ z=t^3 \end{cases}$ 在点

$M(1,1,1)$ 处切线的方向向量

$$\vec{l}=\left(\frac{\mathrm{d}x}{\mathrm{d}t},\frac{\mathrm{d}y}{\mathrm{d}t},\frac{\mathrm{d}z}{\mathrm{d}t}\right)\Bigg|_{t=1}=(1\ ,\ 4\ ,\ 3)$$

$f_x(1,1,1)=2x^2\Big|_{(1,1,1)}=2$ $\qquad f_y(1,1,1)=zy^{z-1}\Big|_{(1,1,1)}=1$

函数沿 $\vec{l}$ 的方向导数 $\qquad f_z(1,1,1)=y^z\ln z\Big|_{(1,1,1)}=0$

$$\left.\frac{\partial f}{\partial l}\right|_M=\left[f_x\cdot\cos\alpha+f_y\cdot\cos\beta+f_z\cdot\cos\gamma\right]\Big|_{(1,1,1)}=\frac{6}{\sqrt{26}}$$

(2) $\operatorname{grad} f\big|_M = (2, 1, 0)$

$$\cos\theta = \frac{\operatorname{grad} f\big|_M \cdot \vec{l}}{\left|\operatorname{grad} f\big|_M\right| \left|\vec{l}\right|} = \frac{\left.\frac{\partial f}{\partial l}\right|_M}{\left|\operatorname{grad} f\big|_M\right|} = \frac{6}{\sqrt{130}}$$

$$\therefore\ \theta = \arccos\frac{6}{\sqrt{130}}$$

七、求 $\ln x+\ln y+3\ln z$ 在球面 $x^2+y^2+z^2=5r^2$ 上的部分最大值， 其中 $x,y,z>0$，且用此结果证明对于正实数 a,b,c 有

$$abc^3 \le 27\left(\frac{a+b+c}{5}\right)^5$$

解 设 $F=\ln x+\ln y+\ln z+\lambda(x^2+y^2+z^2-5r^2)$

$$\begin{cases} F_x=\dfrac{1}{x}-2\lambda x\overset{令}{=}0, \\ F_y=\dfrac{1}{y}-2\lambda y\overset{令}{=}0, \\ F_z=\dfrac{3}{z}-2\lambda z\overset{令}{=}0 \\ x^2+y^2+z^2-5r^2=0 \end{cases}$$

$\Rightarrow x=y=r,\ z=\sqrt{3}r$

$$f(r,r,\sqrt{3}r)=\ln 3\sqrt{3}r^5=\frac{1}{2}\ln 27\cdot r^{10}$$

$$=\frac{1}{2}\ln 27\left(\frac{x^2+y^2+z^2}{5}\right)^5$$

取 $x=\sqrt{a},y=\sqrt{b},z=\sqrt{c}$

即得所要证的不等式！